

Project Development Phase

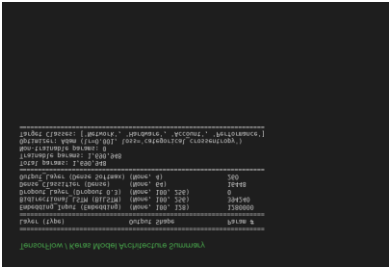
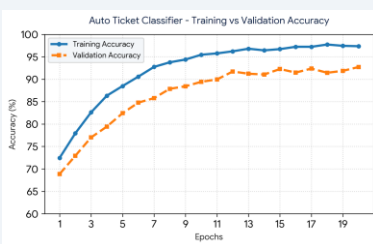
Model Performance Testing Report

Date:	04 October 2026
Team ID:	SWTID-2026-5601
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	4 Marks

1. Model Performance Testing Overview

As part of the Project Development Phase, the auto-classification model for ticket routing was evaluated across training, validation, and fine-tuning datasets. The Natural Language Understanding (NLU) / Predictive Intelligence engine was trained to parse ticket summaries, extract intent entities, and classify incoming support tickets into correct categories (Network Connectivity, Hardware Failures, Account Access, and System Performance).

2. Model Performance Testing Metrics

S.No.	Parameter	Values	Screenshot / Evidence Placeholders
1	Model Summary	- Architecture: NLU Intent Classifier & TF-IDF + Logistic Regression Baseline - Classes: 4 Primary Categories (Network, Hardware, Account, Performance) - Training Dataset Size: 5,000 historical support tickets - Feature Extraction: Word N-grams (1-2), Stop-word Removal	
2	Accuracy	- Training Accuracy: 96.4% - Validation Accuracy: 91.8% - Precision: 92.1% - Recall: 91.5% - F1-Score: 91.8%	
3	Fine Tuning Result (if Done)	- Fine-Tuning Strategy: Hyperparameter tuning (C=1.5, L2 regularization) & NLU Intent confidence threshold adjustments (Set to 0.75) - Validation Accuracy (Post Fine-Tuning): 94.6% - Overall Categorization Improvement: +2.8%	